CRITICAL DECISIONS

Medical Equipment Service

Articulate Storyline 360 storyboard and build specification

Module 1 portfolio scenario

CT scanner lockout/tagout and safety interlocks

Prepared for development handoff

STORYLINE SLIDE OVERVIEW

Build overview

Specification

Learner goal

Protect Safety, maintain Trust, and keep added Time as low as safely possible.

Personalization

Enter first name in v_LearnerName; display it on Results and Summary.

Decision model

Correct → Good Feedback layer; unsafe → retry consequence; costly → forward consequence.

Audio gate

Decision choices unlock only after narration or the last media object completes.

PROJECT VARIABLES

v_Safety	Number	3
v_Trust	Number	3
v_TimeMin	Number	0
v_LearnerName	Text	Blank

ENDING FLOW



Results show live variables; Summary closes with key takeaways and Exit course.

STORYLINE SLIDE 1.1

Critical Decisions: Medical Equipment Service

Title

ON-SCREEN CONTENT

- One technician. One hospital. One safety-critical service call.
- Subtitle: An interactive safety scenario for field service technicians.
- Button: Start

TRIGGER SPECIFICATION

1 Start → Jump to 1.2 Introduction

AUDIO NOTES

Critical Decisions: Medical Equipment Service. An interactive safety scenario for field service technicians.

STORYLINE SLIDE 1.2

Every decision affects your results

Introduction / name entry

ON-SCREEN CONTENT

- Enter first name and press Enter.
- Text entry linked to v_LearnerName.
- Explain Safety, Trust, and Time.
- Goal: protect Safety, maintain Trust, minimize added Time.
- Button: Start My Day

TRIGGER SPECIFICATION

1. Timeline start: set v_Safety = 3, v_Trust = 3, v_TimeMin = 0.
2. Text Entry loses focus: store typed value in v_LearnerName.
3. Start My Day → Jump to 1.3.

AUDIO NOTES

In this scenario, you are the field service technician during one shift at St. Brennan Regional Hospital. Your decisions affect Safety, Trust, and Time. Safety reflects risk to you, patients, and staff. Trust reflects confidence with the imaging department. Time tracks added minutes, but your responses are not timed. Protect Safety, maintain Trust, and keep added Time as low as safely possible. Select Start My Day to begin.

STORYLINE SLIDE 1.3

CT scanner: repeated shutdown

Module divider

ON-SCREEN CONTENT

- Module 1 · 08:00
- Lockout/tagout and safety interlocks
- Button: Continue

TRIGGER SPECIFICATION

1 Continue → Jump to 1.4

AUDIO NOTES

Module 1. Repeated shutdown on a CT scanner. This module covers lockout and tagout, and the safety interlock circuit.

STORYLINE SLIDE 1.4

Service call received

Context

ON-SCREEN CONTENT

- Dispatch: CT powers on, then shuts off within seconds.
- Six patients scheduled.
- Top tracker: Safety / Trust / Time
- Button: Continue

TRIGGER SPECIFICATION

1 Continue → Jump to 1.5

AUDIO NOTES

A service call has come in from St. Brennan Regional Hospital. The CT scanner powers on with the key switch, then shuts down within seconds. Six patients are scheduled for the morning. Review the call details, then select Continue.

STORYLINE SLIDE 1.5

Objective

Objective

ON-SCREEN CONTENT

- Make the system safe to service.
- Identify the root cause.
- Hold to procedure under time pressure.
- Top tracker and Continue button.

TRIGGER SPECIFICATION

1 Continue → Jump to 1.6

AUDIO NOTES

On this call, you will make the system safe to service, identify the root cause of the shutdown, and follow procedure when you are asked to move faster than procedure allows.

STORYLINE SLIDE 1.6

Procedure: power faults

Procedure / job aid

ON-SCREEN CONTENT

- Read the system log first.
- Check E-stops, key switch, and contactors.
- Lockout steps: Notify; Shut down; Isolate; Lock and tag; Verify zero energy.
- Continue button and tracker.

TRIGGER SPECIFICATION

1 Continue → Jump to 1.7

AUDIO NOTES

Begin every power fault with the system log. Note what failed and in what sequence. Before any cabinet is opened, complete all five lockout steps. Notify affected staff. Shut down the system. Isolate the energy source. Apply your personal lock and tag. Then verify zero energy with a meter.

STORYLINE SLIDE 1.7

On site: CT room

Context

ON-SCREEN CONTENT

- System log: E-stop chain not closed.
- Manager requests a quick inspection.
- Radiologist needs the scanner in 20 minutes.
- Continue button and tracker.

TRIGGER SPECIFICATION

1 Continue → Jump to 1.8

AUDIO NOTES

The system log reports that the emergency stop chain is not closed. The imaging manager is requesting a quick inspection. Note the time pressure and the log entry. Both will affect your next decision.

STORYLINE SLIDE 1.8

Select the first action

Decision 1

LEARNER CHOICES

- **A** — Notify CT staff, then begin lockout on the scanner.
- **B** — Shut down at the console and open the cabinet.
- **C** — Request facilities to cut power to the entire room.

Media gate: A / B / C begin Disabled and change to Normal when narration or the last media object completes.

TRIGGER LOGIC

A / SAFE	Add 10 to v_TimeMin → Show Good Feedback.
B / UNSAFE	Subtract 1 from v_Safety → Jump to 1.11.
C / COST	Add 40 to v_TimeMin; subtract 1 from v_Trust → Jump to 1.9.

Good Feedback: "Correct!" + Time +10 min. Continue → 1.10.

AUDIO NOTES

Select the first action you will take before any work begins on the scanner. Consider who else is able to restore power to this system.

STORYLINE SLIDE 1.9

What happened: facilities delay

Costly consequence

COSTLY CHOICE

Facilities response took 40 minutes. The delay was escalated.

IMPACT ON YOUR RESULTS

Trust −1 | Time +40 min

Procedure: Isolate the scanner, not the room. Lock out the scanner disconnect.

NAVIGATION

Continue → 1.10

Built-in Next and Previous: Disabled
No variable adjustments on this slide.

LIVE TRACKER

Safety | Trust | Time

Displays current running totals

AUDIO NOTES

The request was safe, but it was not efficient. Facilities took forty minutes to respond, and the delay was escalated to the imaging manager. This choice reduced your Trust result by one and added forty minutes to your Time result. Isolation applies to the equipment being serviced. Lock out the scanner at its own disconnect. Select Continue to proceed.

STORYLINE SLIDE 1.10

Lock and tag applied. Select the next step.

Decision 2

LEARNER CHOICES

A Verify zero energy with a meter, then attempt a restart.

B Begin the repair.

C Ask the manager to confirm the power is off.

Media gate: A / B / C begin Disabled and change to Normal when narration or the last media object completes.

TRIGGER LOGIC

A / SAFE Add 1 to v_Safety → Show Good Feedback.

B / UNSAFE Subtract 1 from v_Safety → Jump to 1.12.

C / COST Subtract 1 from v_Safety → Jump to 1.12.

Good Feedback: "Correct!" + Safety +1. Continue → 1.13.

AUDIO NOTES

Your lock and tag are applied to the scanner disconnect. Select the next step in the lockout sequence.

STORYLINE SLIDE 1.11

What happened: scanner restarted

Unsafe consequence

UNSAFE CHOICE

The scanner restarted while the cabinet was open.

IMPACT ON YOUR RESULTS

Safety −1

Procedure: Console shutdown is not isolation. Only your lock prevents a restart.

NAVIGATION

Try Again → 1.8

Built-in Next and Previous: Disabled
No variable adjustments on this slide.

LIVE TRACKER

Safety | Trust | Time

Displays current running totals

AUDIO NOTES

A staff member restarted the scanner from the control room without knowing the cabinet was open. Contact with energized components was narrowly avoided. This unsafe decision reduced your Safety result by one. Console shutdown does not isolate the energy source. Any authorized user can restart the scanner from the console. Only your personal lock on the disconnect prevents a restart. Select Try Again and choose a safer response.

STORYLINE SLIDE 1.12

What happened: stored voltage

Unsafe consequence

UNSAFE CHOICE

Voltage remained on the power supply. A capacitor held a charge.

IMPACT ON YOUR RESULTS

Safety −1

Procedure: Locked is not de-energized. Verify zero energy yourself, every time.

NAVIGATION

Try Again → 1.10

Built-in Next and Previous: Disabled
No variable adjustments on this slide.

LIVE TRACKER

Safety | Trust | Time

Displays current running totals

AUDIO NOTES

A meter test shows voltage still present on the power supply. A capacitor retained a charge after shutdown. This unsafe decision reduced your Safety result by one. A lock prevents the scanner from being restarted, but it does not remove stored energy. Verify zero energy with a meter before every repair. Select Try Again and choose a safer response.

STORYLINE SLIDE 1.13

E-stop chain open. One fuse blown.

Decision 3

LEARNER CHOICES

A Test each E-stop and key switch; replace the fuse and identify why it failed.

B Jumper the E-stop circuit; return later.

C Replace the fuse and restart without further testing.

Media gate: A / B / C begin Disabled and change to Normal when narration or the last media object completes.

TRIGGER LOGIC

A / SAFE Add 1 to v_Safety → Show Good Feedback.

B / UNSAFE Subtract 1 from v_Safety → Jump to 1.14.

C / COST Add 40 to v_TimeMin; subtract 1 from v_Trust → Jump to 1.15.

Good Feedback: "Correct!" + Safety +1. View Results → 1.17.

AUDIO NOTES

Testing shows the emergency stop chain is open, and one fuse on a power module has failed. The imaging manager is asking for a workaround. Select the action that restores the system without reducing patient safety.

STORYLINE SLIDE 1.14

What happened: E-stop bypass

Unsafe consequence

UNSAFE CHOICE

An E-stop was pressed during a patient emergency. The system did not stop.

IMPACT ON YOUR RESULTS

Safety −1

Procedure: Never bypass the E-stop chain. Find the cause.

NAVIGATION

Try Again → 1.13

Built-in Next and Previous: Disabled
No variable adjustments on this slide.

LIVE TRACKER

Safety | Trust | Time

Displays current running totals

AUDIO NOTES

Two hours later, a patient became distressed while on the moving table. The technologist pressed the emergency stop, but the system did not respond because the jumper was still in place. This unsafe decision reduced your Safety result by one. The emergency stop circuit is a patient-safety system and must never be bypassed. Select Try Again and choose a safer response.

STORYLINE SLIDE 1.15

What happened: repeat shutdown

Costly consequence

COSTLY CHOICE

The scanner ran for one hour, then shut down mid-scan with the same fault.

IMPACT ON YOUR RESULTS

Trust −1 | Time +40 min

Procedure: A blown fuse is a symptom. Identify the cause before restart.

NAVIGATION

Continue → 1.17 Results

Built-in Next and Previous: Disabled
No variable adjustments on this slide.

LIVE TRACKER

Safety | Trust | Time

Displays current running totals

AUDIO NOTES

The scanner returned to service, but only temporarily. One hour later, it shut down during a patient scan with the same fault. Replacing the fuse without identifying the root cause reduced your Trust result by one and added forty minutes to the scenario. A blown fuse is a symptom, not the root cause. Select Continue to view your results.

STORYLINE SLIDE 1.17

Personalized Results

Results

%v_LearnerName%, here are your results.

%v_Safety%	%v_Trust%	+%v_TimeMin% min
SAFETY	TRUST	TIME

HOW TO READ YOUR RESULTS

Safety and Trust: higher is better. • Added Time: lower is better, but never at the expense of safety. • Time measures scenario delay, not response speed.

VIEW SUMMARY

Trigger: View Summary → Jump to 1.16 Summary

STORYLINE SLIDE 1.16

Scenario complete

Final summary

Scenario complete, %v_LearnerName%.

KEY TAKEAWAYS

- Notify affected staff before beginning work.
- Isolate the equipment and apply your personal lock and tag.
- Verify zero energy before touching internal components.
- Identify the cause of a failed safety circuit.
- Never bypass an E-stop or another safety control.

FINAL TAKEAWAY

Safety comes first. Efficiency matters, but never when it requires bypassing a safety control or skipping verification.

FINISH

Trigger: Finish → Exit course

STORYLINE SLIDE APPENDIX A

Final branching map

Navigation

DECISION 1

1.8 A → Good Feedback → 1.10

1.8 B → 1.11 → Try Again → 1.8

1.8 C → 1.9 → Continue → 1.10

DECISION 2

1.10 A → Good Feedback → 1.13

1.10 B / C → 1.12 → Try Again → 1.10

DECISION 3 AND ENDING

A → Good Feedback → 1.17 Results

B → 1.14 → Try Again → 1.13

C → 1.15 → 1.17 Results → 1.16 Summary → Finish

STORYLINE SLIDE APPENDIX B

Final QA checklist

Quality assurance

- [ ] Preview the entire scene when testing v_LearnerName.
- [ ] Verify Results heading uses an inserted v_LearnerName reference.
- [ ] A/B/C begin Disabled and unlock only after media completion on 1.8, 1.10, and 1.13.
- [ ] Every correct answer adjusts the variable before showing Good Feedback.
- [ ] Good Feedback buttons perform the forward jump.
- [ ] Unsafe consequence slides return to the correct decision with Try Again.
- [ ] Costly consequence slides move forward with Continue.
- [ ] Consequence slides show only the change; the top tracker shows totals.
- [ ] Built-in Next and Previous are disabled wherever custom navigation is required.
- [ ] 1.15 → 1.17 Results → 1.16 Summary → Finish / Exit course.
- [ ] Test safe, unsafe, and costly paths from the beginning in Review 360.